1 **(a)** State two differences between *random* and *systematic* errors.

.....

.....

..... [2]

(b) Fig. 1.1 shows a simple pendulum consisting of a mass m attached to a light inextensible string of length L. The pendulum is secured to a fixed point and made to undergo oscillations when displaced sideways by a small angle θ.

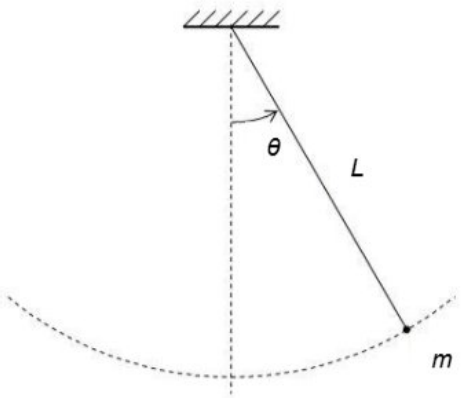


Fig. 1.1 (not to scale)

The period of oscillation T can be found by the following equation:

$$T^2 g = 4\pi^2 L$$

where g is the acceleration of free fall.

(i) Given $L = 50.0 \pm 0.2$ cm; $g = 9.8 \pm 0.1$ m s^{-2}, find T with its associated uncertainty.

T = ... ± ..... s [3]

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(ii) When determining period T directly using a stopwatch, it is a good practice to take a large number of oscillations.

Explain how this makes the result more reliable.

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..... [1]

[Total: 6]